



Influence of Non-Fourier Heat Flux on the Hybrid Nanofluid Flow over a Permeable Sphere When Lorentz Force, Soret and Dufour Effects Are Significant: An Artificial Neural Network Model

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Abstract

Heat transmission in systems or thermal devices relies on the thermal conductivity of nanofluids. Thermal conductivity (32%) of the hybrid nanofluid (i.e., ethylene glycol conveying zinc and titania nanoparticles) is higher than that (13%) of the ordinary nanofluid (i.e., ethylene glycol conveying zinc nanoparticles), according to experimental data. The present study investigated the steady, laminar flow of a hybrid nanofluid over a sphere with thermal radiation, non-Fourier heat flux, and cross-diffusion effects. The Keller box scheme with the combination of finite difference procedure is opted to solve the governing equations by transforming them into a nonlinear and non-dimensional system of partial differential equations. Outcomes (such as the variation in the temperature profile due to the Dufour number) are presented in the form of bar plots and plots in two cases, i.e., $EG + ZnO + TiO_2$ and $EG + ZnO$. An in-built function, 'nnstart' (ANN), in MATLAB is used to validate the data of the engineering parameters, including the skin friction coefficient. It is noticed that the friction factor decreases by 2.5% (in the case of hybrid nanofluid) and 2.19% (in the case of mono nanofluid) when the value of the magnetic field parameter ranges from 0 to 3. It is observed that the Nusselt number decreases by 39.7% (in the case of hybrid nanofluid) and 38.5% (in case of mono nanofluid) when the value of the Dufour parameter ranges from 0 to 3. It is detected that the Sherwood number decreases by 2.7% (in the case of hybrid nanofluid) and 2.97% (in the case of mono nanofluid) when the value of the Soret parameter ranges from 0 to 3. Furthermore, it is observed that the fluid temperature decreases with the rise in the thermal relaxation parameter.

Keywords Hybrid nanofluid · Thermal radiation · Soret and Dufour · Chemical reaction · Artificial neural network · Keller box

List of symbols

u, v The components of velocity in x, y directions
 μ Viscosity of the fluid ($kgm^{-1}s^{-1}$)

Extended author information available on the last page of the article

ν	Kinematic viscosity
ρ	Density of the fluid (kgm^{-3})
Pr	Prandtl number
g	Acceleration due to gravity
β_T	Thermal expansion coefficient
T	Temperature of fluid (dimensional) (K)
Sc	Schmidt number
k^*	Mean absorption coefficient
f'	Non-dimensional velocity
C	Concentration of fluid (dimensional) (molm^{-3})
ψ	Stream function
f	The non-dimensional stream function
C_p	Specific heat capacity
k	Thermal conductivity ($\text{m}^{-1}\text{K}^{-1}$)
η	Similarity variable
δ	Dimensionless thermal relaxation time parameter
δ^*	Thermal relaxation time of heat flux
ξ	Dimensionless coordinate
β_C	Diffusion expansion coefficient
θ	Non-dimensional temperature
k_T	Thermal mean temperature
T_m	Fluid mean temperature
C_s	Concentration susceptibility
K	Permeability of porous medium
Φ	Non-dimensional concentration of fluid
E_{ck}	Eckert number
Gr	Local thermal Grashof number
Gc	Local diffusion Grashof number
λ	Buoyancy ratio parameter
M	Magnetic field parameter
D_m	Molecular diffusivity (m^2s^{-1})
Du	Dufour number
Sr	Soret number
σ	Electrical conductivity
k_1	Chemical reaction rate
Kr	Dimensionless chemical reaction parameter

Subscripts

f	Fluid
nf	Nanofluid
hnf	Hybrid Nanofluid

Background Information

Cattaneo-Christov heat flux, also known as the Cattaneo-Vernotte equation or the hyperbolic heat conduction equation, is a modified form of the heat conduction equation that accounts for finite heat propagation speeds. It was developed to address some of the limitations of the classical Fourier heat conduction equation, which assumes instantaneous heat transfer throughout a material. In the classical Fourier heat conduction equation, changes in temperature propagate instantly throughout a material, implying infinite heat propagation speed. However, in reality, heat transfer does not occur instantaneously, and there is a finite speed at which temperature disturbances propagate through a material. This finite speed is particularly important when dealing with materials with high thermal conductivity or when studying heat transfer at extreme time scales. This modified heat conduction equation is often used in the study of heat transfer in various fields, including fluid dynamics, solid-state physics, and materials science. It is particularly useful when analyzing heat conduction in materials with short thermal relaxation times, such as certain metals and highly conductive materials. After the partial time derivation in Maxwell-Cattaneo's law, Christov [1], developed Oldroyd's upper-convected derivative to maintain the model's material invariance (which is Cattaneo-Christov law). Later, Straughan [2] emphasized that the Christov heat flux rule, which is correctly invariant, has a significant impact on thermal convection. In the slip regime, Han et al. [3] investigated the Maxwell fluid flow via an unfolded plate using the Cattaneo-Christov model, which was included in the energy equation. It was concluded that the fluid temperature decreases with an increase in the heat flux relaxation time parameter. Analysis of heat transfer in a vertical MHD flow issue (with the CCHF model) was carried out by Li et al. [4] using the Bvp4c approach. It was found that as the thermal relaxation time parameter was increased, the heat transfer rate increased. Examination of the Casson fluid flow across an uneven elongating sheet with varying viscosity is done by Malik et al. [5]. It turns out that the temperature rises more quickly in the Fourier's law case than in the CCHF model. A similar conclusion was reached by Dogonchi and Ganji [6, 7] in their study of the flow of nanofluids over distinct geometries with the CCHF model. Using the same model, Ramandevi et al. [8] discussed the effect of irregular heat sink/source on distinct fluid flows via an extending sheet. The oblique nanofluid (Cu-Water) flow via a porous shrinking/extending surface towards a stagnation point was numerically explored by Li et al. [9]. For shrinking, they observed that the fluid was able to transmit more heat per unit volume than for stretching with a bigger suction parameter. With the CCHF model, Rana and Nawaz [10] studied the effect of Ohmic heating on the dissipative flow of Sutterby nanofluid. Thermal boundary layer thickness seemed to be smaller for Al_2O_3 and Sutterby nanofluid blend than for CuO and Sutterby nanofluid combination when the thermal relaxation time parameter was taken into account. Water-based CNT fluid flow amongst coaxial revolving discs with a CCHF model was analyzed by Bhattacharyya et al. [11] using OHAM. Several scholars [12–16] have recently looked at the CCHF model and examined various fluid flows in various geometries, including spinning discs.

The Soret and Dufour effects are two phenomena that describe the coupling of the transfer of heat and mass in a fluid flow. The Soret effect is the propensity of a solute to diffuse in the direction of a temperature gradient, and the Dufour effect is the propensity of heat to diffuse in the direction of a concentration gradient. Understanding and accounting for these effects are essential for accurate modeling and prediction of behavior in systems where

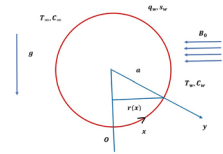
temperature gradients and concentration gradients coexist, such as in combustion, environmental transport, and microfluidics. Due to this reason, these effects are important in fluid mechanics for the huge range of concentration and temperature gradients and their active part in macroscopically observable physical phenomena (Moorthy and Senthilvadivu [17] and Reddy and Chamkha [18]). In their study of the MHD fluid flow through a flat plate with Dufour and Soret effects, Mahdy and Ahmed [19] discovered that the temperature of the fluid rose as the Dufour number rose. Irreversibility analysis in power-law fluid flow within a cavity was explored by Kefayati [20]. An increment in mass transfer with a rise in the Soret parameter was noted. The nonlinear radiative flow of Maxwell fluid via a bidirectional elongating surface was studied by Ramzan et al. [21] using *bvp4c*, and it was noticed that a rise in fluid temperature corresponded to a rise in the ratio parameter. Using the Dufour and Soret effects, Farooq et al. [22] and Shojaei et al. [23] studied a variety of non-Newtonian fluid flows on stretched surfaces. The outcome of the study indicates that when the Dufour number augments, the heat transmission rate is decreased. Idowu and Falodun [24] and Jha and Gambo [25] use the same parameters to conduct an analytical analysis of distinct (steady/unsteady) MHD fluid flows across a vertical plate. It was discovered that increasing the Soret number results in an improvement in the shear rate. Rasool et al. [26] described the non-Darcy flow of nanofluid across an elongating sheet using the same effects. It is discovered that an increment in the Forchheimer number decreases fluid velocity. Various writers [27–33] have recently explored different fluid flows using different geometry and the same effects.

The effect of magnetic fields on fluid flow holds significant importance in both scientific research and industrial applications. Magnetohydrodynamics (MHD) plays a crucial role in understanding plasma behavior in astrophysics, controlling molten metal flows in metallurgical processes, and enhancing heat transfer in advanced nuclear reactors. In engineering, MHD principles are applied in electromagnetic pumps for liquid metal cooling systems, magnetic drug targeting in biomedical applications, and the design of fusion reactors. The study of magnetic field effects on hybrid nanofluids, in particular, offers potential for optimizing thermal management systems in electronics cooling, solar energy collectors, and aerospace thermal protection systems. Shilpa and Leela [34] conducted a numerical study using the Galerkin Finite Element Method (GFEM) to analyze heat and mass transfer in Jeffrey, Maxwell, and Oldroyd-B nanofluids within a vertical annulus under an induced magnetic field and non-uniform heat source/sink. The study revealed that the Jeffrey nanofluid model outperformed Maxwell and Oldroyd-B models in heat and mass transfer rates, with Lorentz force reducing velocity and IMF boundary layers while temperature distribution increased with higher Brownian motion. Shilpa and Leela [35] investigated magnetohydrodynamic mixed convective flow in an inclined cylindrical annulus, incorporating heat generation/absorption and chemical reactions, using an implicit finite difference method and an artificial neural network model. The study demonstrated that increasing the inclination angle enhances thermal and concentration profiles, while higher chemical reaction rates thicken the concentration boundary layer. Later, various researchers [36–41] discussed the influence of magnetic field on different fluid flows over distinct geometries.

Increasing effects of heat and mass flux due to concentration and temperature gradients on the hybrid nanofluid flow across a sphere using the Cattaneo-Christov heat flux model under the convective boundary conditions have not been studied in the literature. When Lorentz force, thermal radiation, viscous dissipation, heat flux due to concentration gradi-

Table 1 Specific quantities for base fluid and nanomaterials (Jamshed and Aziz [44])

S.No.	Properties	Ethylene glycol (EG) (f)	ZnO (φ_1)	TiO ₂ (φ_2)
1.	σ (sm ⁻¹)	0.0000055	0.01	2,380,000
2.	k (W(mK) ⁻¹)	0.252	13	8.953
3.	ρ (kgm ⁻¹)	1114	5660	4250
4.	C_p (J(kgK) ⁻¹)	2415	495.2	686.2

Fig. 1 Graphical illustration of the dynamics

ent, and mass flux due to thermal gradient are significant, nothing is known about the self-similar flow of chemically reactive ethylene glycol carrying zinc and titania nanoparticles through the permeable sphere. Hence, this study was designed to

1. compare the skin friction coefficient, Nusselt number (transfer of heat at the wall), and Sherwood number (transfer of mass at the wall) in the motion of ethylene glycol with zinc and titania nanoparticles versus ethylene glycol with zinc nanoparticles at various levels of Lorentz force, volume fraction of nanoparticles, Darcy number, viscous dissipation parameter, Dufour, Soret, and chemical reaction.
2. reveal the impact of volume fraction of nanoparticles on the velocity profile.
3. determine the effects of Dufour parameter and thermal relaxation parameters on the temperature profile.
4. observe the impact of Soret parameter on the concentration of the hybrid nanofluid (i.e., ethylene glycol with zinc and titania nanoparticles) and mono nanofluid (i.e., ethylene glycol with zinc nanoparticles).

Research Methodology

The goal of this work is to discover the influences of Soret and Dufour parameters on the flow of a hybrid nanofluid ($EG + ZnO + TiO_2$) across a convectively heated sphere along with viscous dissipation as well as chemical reaction. The energy equation is extended using the heat flux model of Cattaneo-Christov. Normal to the surface, a magnetic field with a strength of B_0 is applied. The base fluid and nanomaterials' thermophysical characteristics are shown in Table 1. Fluid flow is only taken into account approaching the point of stagnation. We also anticipated the induced magnetic field to be discarded (Fig. 1).

In agreement with Alam et al. [42] and Rao et al. [43], the governing equations for this problem transport phenomenon over a permeable sphere are:

$$\frac{\partial(rv)}{\partial y} + \frac{\partial(ru)}{\partial x} = 0 \quad (1)$$

$$v \frac{\partial}{\partial y} (u) + u \frac{\partial}{\partial x} (u) = v_{hnf} \frac{\partial}{\partial y} \left(\frac{\partial}{\partial y} (u) \right) + g (T - T_{\infty}) \beta_T \sin \left(\frac{x}{a} \right) - \frac{v_{hnf}}{K} u - B_0^2 \frac{\sigma_{hnf}}{\rho_{hnf}} u + g (C - C_{\infty}) \beta_C \sin \left(\frac{x}{a} \right) \quad (2)$$

$$\begin{aligned} v \frac{\partial}{\partial y} (T) + u \frac{\partial}{\partial x} (T) + \delta^* \left[\begin{aligned} &u^2 \frac{\partial}{\partial x} \left(\frac{\partial}{\partial x} (T) \right) + v^2 \frac{\partial}{\partial y} \left(\frac{\partial}{\partial y} (T) \right) + 2uv \frac{\partial}{\partial x} \left(\frac{\partial}{\partial y} (T) \right) \\ &+ \frac{\partial T}{\partial x} \left[u \frac{\partial}{\partial x} (u) + v \frac{\partial}{\partial y} (u) \right] + \frac{\partial T}{\partial y} \left[u \frac{\partial}{\partial x} (v) + v \frac{\partial}{\partial y} (v) \right] \end{aligned} \right] \\ = \frac{k_{hnf}}{(\rho C_p)_{hnf}} \frac{\partial}{\partial y} \left(\frac{\partial}{\partial y} (T) \right) + \frac{16\sigma^* T_{\infty}^3}{3k^*} \frac{1}{(\rho C_p)_{hnf}} \frac{\partial}{\partial y} \left(\frac{\partial}{\partial y} (T) \right) \\ + \frac{\mu_{hnf}}{(\rho C_p)_{hnf}} \left(\frac{\partial}{\partial y} (u) \right) \left(\frac{\partial}{\partial y} (u) \right) + \frac{D_m k_T}{c_s C_p} \frac{\partial}{\partial y} \left(\frac{\partial}{\partial y} (C) \right) \end{aligned} \quad (3)$$

$$v \frac{\partial}{\partial y} (C) + u \frac{\partial}{\partial x} (C) = D_m \frac{\partial}{\partial y} \left(\frac{\partial}{\partial y} (C) \right) + \frac{D_m k_T}{T_m} \frac{\partial}{\partial y} \left(\frac{\partial}{\partial y} (T) \right) - (C - C_{\infty}) k_1 \quad (4)$$

with the boundary conditions

$$\left. \begin{aligned} v(x, 0) = u(x, 0) = 0, T(x, 0) = T_w, C(x, 0) = C_w \\ u(x, \infty) \rightarrow 0, T(x, \infty) \rightarrow T_{\infty}, C(x, \infty) \rightarrow C_{\infty}. \end{aligned} \right\}. \quad (5)$$

Hybrid Nanofluid Thermophysical Characteristics

$$\left. \begin{aligned} (\rho C_p)_{hnf} &= \left[(1 - \phi_1) (\rho C_p)_f + \phi_1 (\rho C_p)_1 \right] (1 - \phi_2) + (\rho C_p)_2 \phi_2, \\ \sigma_{nf} &= \sigma_f \times \frac{\sigma_1 + 2\sigma_f - 2\sigma_f \phi_1 + 2\sigma_1 \phi_1}{\sigma_1 + 2\sigma_f + \sigma_f \phi_1 - \sigma_1 \phi_1}, k_{hnf} = k_{nf} \times \frac{k_2 + 2k_{nf} - 2k_{nf} \phi_2 + 2k_2 \phi_2}{k_2 + 2k_{nf} + k_{nf} \phi_2 - k_2 \phi_2}, \\ \rho_{hnf} &= [(1 - \phi_1) \rho_f + \phi_1 \rho_1] (1 - \phi_2) + \rho_2 \phi_2, \\ \mu_{hnf} &= \frac{\mu_f}{(1 - \phi_1)^{2.5} (1 - \phi_2)^{2.5}}, k_{nf} = k_f \times \frac{k_1 + 2k_f - 2k_f \phi_1 + 2k_1 \phi_1}{k_1 + 2k_f + k_f \phi_1 - k_1 \phi_1}, \\ \sigma_{hnf} &= \sigma_{nf} \times \frac{\sigma_2 + 2\sigma_{nf} - 2\sigma_{nf} \phi_2 + 2\sigma_2 \phi_2}{\sigma_2 + 2\sigma_{nf} + \sigma_{nf} \phi_2 - \sigma_2 \phi_2}. \end{aligned} \right\}$$

Note that the radial distance between the harmonic axis along with the sphere surface corresponds to the radius of sphere a is $r = a \sin \left(\frac{x}{a} \right)$. The controlling equations are renovated as a series of ODEs using the similarity transformations shown below (Alam et al. [42]):

$$\left. \begin{aligned} \eta &= \frac{y}{a} (Gr)^{1/4}, \psi = v \xi (Gr)^{1/4} f(\xi, \eta), \xi = \frac{x}{a}, \\ T &= T_{\infty} + (T_w - T_{\infty}) \theta(\xi, \eta), \Phi(\xi, \eta) = \frac{C - C_{\infty}}{C_w - C_{\infty}} \end{aligned} \right\} \quad (6)$$

It is believed that u, v are velocity components that are given by

$$u = \frac{1}{r} \frac{\partial}{\partial \eta} (\psi), v = -\frac{1}{r} \frac{\partial}{\partial \xi} (\psi), \quad (7)$$

so that the continuity equation can be satisfied. Equations (2–4) and boundary conditions (5) are transmuted using (6) and (7) as below:

$$\begin{aligned} & \frac{1}{A_1 A_2} f''' + \left(f f'' (1 + \xi \cot \xi) - \xi \left(f' \frac{\partial}{\partial \xi} (f') - f'' \frac{\partial}{\partial \xi} (f) \right) - f'^2 \right) \\ & - \frac{1}{Da} f' - \frac{A_3 M}{A_2} f' + \lambda \frac{\sin \xi}{\xi} \Phi + \frac{\sin \xi}{\xi} \theta = 0 \end{aligned} \quad (8)$$

$$\begin{aligned} & \left(\frac{A_4}{A_5 \text{Pr}} + \frac{Ra}{A_4 \text{Pr}} \right) \theta'' + \frac{1}{A_1 A_5} E_{ck} f''^2 + Du \Phi'' + \left(\frac{(1 + \xi \cot \xi) f \theta'}{-\xi \left(f' \frac{\partial}{\partial \xi} (\theta) - \theta' \frac{\partial}{\partial \xi} (f) \right)} \right) = \\ & \delta \left[\begin{aligned} & \xi^2 f'^2 \frac{\partial}{\partial \xi} \left(\frac{\partial}{\partial \xi} (\theta) \right) + f^2 \theta'' + \xi^2 \left(\frac{\partial}{\partial \xi} (f) \right) \left(\frac{\partial}{\partial \xi} (f) \right) \theta'' + 2 \frac{\partial}{\partial \xi} (f) \xi f \theta'' \\ & - 2 \xi \frac{\partial}{\partial \xi} \left(\frac{\partial}{\partial \eta} (\theta) \right) f f' + 2 \xi^2 f' \frac{\partial}{\partial \xi} \left(\frac{\partial}{\partial \eta} (\theta) \right) \frac{\partial}{\partial \xi} (f) + \xi^2 f' \frac{\partial \theta}{\partial \xi} \frac{\partial}{\partial \xi} \left(\frac{\partial}{\partial \eta} (f) \right) \\ & + \xi \frac{\partial}{\partial \xi} (\theta) f'^2 - \xi^2 f'^2 \frac{\partial}{\partial \xi} (\theta) \cot \xi - \xi \frac{\partial}{\partial \xi} (\theta) f f'' - \xi^2 \frac{\partial}{\partial \xi} (\theta) \frac{\partial}{\partial \xi} (f) f'' - 2 \xi \frac{\partial}{\partial \xi} (f) f' \theta' \\ & - \xi^2 \frac{\partial}{\partial \xi} \left(\frac{\partial}{\partial \xi} (f) \right) f' \theta' + \xi \cot \xi \theta' f f' + \xi^2 \cot \xi \frac{\partial}{\partial \xi} (f) f' \theta' + f f' \theta' \\ & - \xi^2 \frac{\partial}{\partial \xi} (f) f' \theta' \cot \xi + f f' \theta' + \xi \frac{\partial}{\partial \eta} \left(\frac{\partial}{\partial \xi} (f) \right) f \theta' + \xi \frac{\partial}{\partial \xi} (f) f' \theta' + \xi^2 \frac{\partial}{\partial \eta} \left(\frac{\partial}{\partial \xi} (f) \right) \frac{\partial}{\partial \xi} (f) \theta' \end{aligned} \right] \quad (9) \end{aligned}$$

$$\left(f' \frac{\partial}{\partial \xi} (\Phi) - \Phi' \frac{\partial}{\partial \xi} (f) \right) \xi - f \Phi' (\xi \cot \xi + 1) = \frac{1}{Sc} \Phi'' - Kr \Phi + Sr \theta'' \quad (10)$$

$$\left. \begin{aligned} f(\xi, 0) &= 0, f'(\xi, 0) = 0, \theta(\xi, 0) = 1, \Phi(\xi, 0) = 1, \\ f'(\xi, \infty) &\rightarrow 0, \theta(\xi, \infty) \rightarrow 0, \Phi(\xi, \infty) \rightarrow 0 \end{aligned} \right\} \quad (11)$$

Here ' (prime) stands for the derivative with respect to η .

Because the flow is close to the sphere's stagnation point, we can assume that $\xi \approx 0$. Equations (8–10) with the conditions (11) may be rewritten with this condition as:

$$\frac{1}{A_1 A_2} f''' + \theta - \frac{A_3 M}{A_1} f' - \frac{1}{A_1 A_2} \frac{1}{Da} f' + \lambda \Phi + 2 f f'' = 0 \quad (12)$$

$$\left(\frac{A_4}{A_5 \text{Pr}} + \frac{Ra}{A_5 \text{Pr}} \right) \theta'' + 2 f \theta' + \frac{1}{A_2 A_5} E_{ck} f''^2 + Du \Phi'' - \delta [f^2 \theta'' + 2 f f' \theta'] = 0 \quad (13)$$

$$\frac{1}{Sc} \Phi'' + 2 f \Phi' + Sr \theta'' - Kr \Phi = 0 \quad (14)$$

$$\left. \begin{aligned} f(0, 0) &= 0, f'(0, 0) = 0, \theta(0, 0) = 1, \Phi(0, 0) = 1, \\ f'(0, \infty) &\rightarrow 0, \theta(0, \infty) \rightarrow 0, \Phi(0, \infty) \rightarrow 0. \end{aligned} \right\} \quad (15)$$

Here

$$\left. \begin{aligned} A_1 &= (1 - \phi_2) \left[(1 - \phi_1) + \phi_1 \frac{\rho_1}{\rho_f} \right] + \phi_2 \frac{\rho_2}{\rho_f}, A_2 = (1 - \phi_1)^{2.5} (1 - \phi_2)^{2.5}, \\ A_{31} &= \frac{\sigma_1 + 2\sigma_f - 2\phi_1 (\sigma_f - \sigma_1)}{\sigma_1 + 2\sigma_f + \phi_1 (\sigma_f - \sigma_1)}, A_{41} = \frac{k_1 + 2k_f - 2\phi_1 (k_f - k_1)}{k_1 + 2k_f + \phi_1 (k_f - k_1)}, \\ A_3 &= \frac{\sigma_2 + 2A_{31}\sigma_f - 2\phi_2 (A_{31}\sigma_f - \sigma_2)}{\sigma_2 + 2A_{31}\sigma_f + \phi_2 (A_{31}\sigma_f - \sigma_2)} A_{31}, A_4 = \frac{k_2 + 2A_{41}k_f - 2\phi_2 (A_{41}k_f - k_2)}{k_2 + 2A_{41}k_f + \phi_2 (A_{41}k_f - k_2)} A_{41}, \\ A_5 &= (1 - \phi_2) \left[(1 - \phi_1) + \phi_1 \frac{(\rho C_p)_1}{(\rho C_p)_f} \right] + \phi_2 \frac{(\rho C_p)_2}{(\rho C_p)_f}. \end{aligned} \right\}$$

and

$$\left. \begin{aligned} M &= \frac{a^2 \sigma_f B_0^2}{\rho \nu Gr^{0.5}}, Pr = \frac{\mu C_p}{k_f}, E_{ck} = \frac{v^2 Gr \xi^2}{a^2 C_p (T_w - T_\infty)}, \lambda = \frac{Gc}{Gr}, Gc = \frac{g \beta_c (C_w - C_\infty) a^3}{v^2}, \\ Gr &= \frac{g \beta_T (T_w - T_\infty) a^3}{v^2}, Sc = \frac{\nu}{D_m}, Ra = \frac{16 \sigma^* T_\infty^3}{3 k^* k}, Da = \frac{\kappa Gr^{0.5}}{a^2}, \\ Du &= \frac{D_m k_T (C_w - C_\infty)}{C_p c_s (T_w - T_\infty)}, Sr = \frac{D_m k_T (T_w - T_\infty)}{v T_m (C_w - C_\infty)}, Kr = \frac{k_1 a^2}{v Gr^{0.5}}, \delta = \frac{\delta^* \nu}{r a^2} Gr^{0.5} \end{aligned} \right\}.$$

Skin friction coefficient is defined as:

$$Cf = \frac{Gr^{-\frac{3}{4}} a^2}{\mu v} \tau_w \quad (16)$$

and $\tau_w = \mu_{hnf} \left(\frac{\partial u}{\partial y} \right) \Big|_{y=0}$

Using Eqs. (6) and (7), the term in Eq. (16) can be reduced to

$$Cf = \frac{1}{A_1} \xi f''(\xi, 0).$$

Nusselt and Sherwood numbers are defined as:

$$Nu = \frac{q_w}{k_f} \frac{a}{(T_w - T_\infty)} Gr^{-\frac{1}{4}}, Sh = \frac{s_w}{D_m} \frac{a}{(C_w - C_\infty)} Gr^{-\frac{1}{4}}. \quad (17)$$

Here, the wall heat flux $q_w = -\frac{\partial T}{\partial y} \left(k_{hnf} + \frac{16 \sigma^* T_\infty^3}{3 k^*} \right) \Big|_{y=0}$, and wall mass flux $s_w = -D_m \frac{\partial C}{\partial y} \Big|_{y=0}$.

Utilizing (6) and (7), quantities mentioned in (17) can be redrafted as below:

$$Nu = -A_4 (1 + Ra) \theta'(\xi, 0) \text{ and } Sh = -\Phi'(\xi, 0).$$

Numerical Procedure

The Eqs. (8–10), along with the conditions (11), are solved by means of the Keller box scheme with a combination of finite difference method. The Keller-box method is particularly suited for solving boundary layer problems due to its inherent stability and second-order accuracy, which arise from its implicit finite-difference formulation and box-scheme discretization. Its use is justified here because it efficiently handles stiff differential equations, such as those governing fluid flow and heat transfer, by transforming them into a system of first-order equations before applying a Newton-based iterative solver. However, limitations include increased computational complexity due to the coupled nonlinear algebraic system requiring iterative methods (e.g., Newton-Raphson) for resolution. Convergence relies on appropriate initial guesses and step sizes, with criteria typically based on the relative error between successive iterations falling below a small tolerance (e.g., 10^{-5}). The method may struggle with highly nonlinear or singular problems unless adaptive meshing or under-relaxation is employed. Clearer documentation of these aspects—step size selection, error thresholds, and handling of nonlinearities—would strengthen reproducibility.

Validation of Present Results

Validation is carried out between our results (wall temperature and friction coefficient) and previously reported results for the case ($\phi_1 = \phi_2 = 0$; a Newtonian fluid). We found a reasonable concordance (see Table 2).

Discussion of the Results

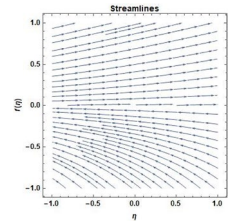
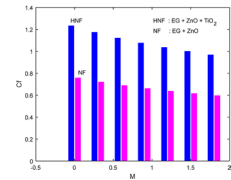
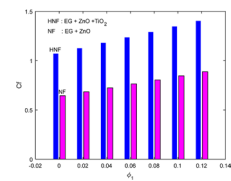
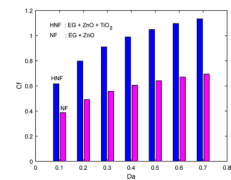
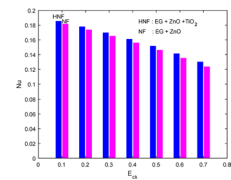
The numerical results initiated at the beneath stagnation point of the sphere ($\xi \approx 0$), due to a primary profile from those Eqs. (12–14) together with the conditions (15) and continued about the sphere at the point $\xi \approx \frac{2\pi}{3}$ (see Molla et al. [46]). The results described for two distinct types of nanofluids which are mono nanofluid ($EG + ZnO$) as well as for hybrid nanofluid ($EG + ZnO + TiO_2$). Figure 2 displayed the streamlines of the current study (in case of hybrid nanofluid flow).

Engineering Parameters of Interest Including Skin Friction Coefficient

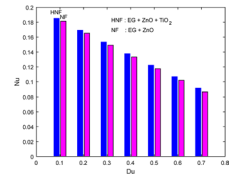
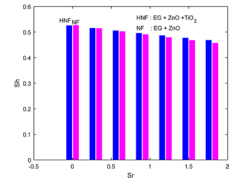
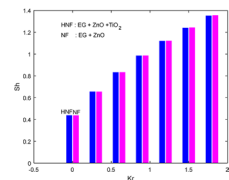
From Figs. 3, 4 and 5, it is clear that there is a decrement in the skin friction coefficient with the rise in the magnetic field parameter, but the same rises with the rise in both volume fraction of nanoparticles of zinc and Darcy number. The rise in the volume fraction of nanoparticles increases the skin friction coefficient due to enhanced viscous drag and interparticle

Table 2 Validation of present outcomes by comparing with previous studies discussed under the more specialized situations

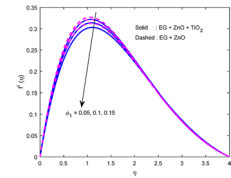
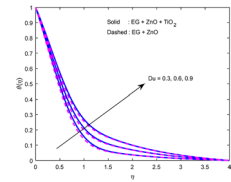
Pr	Wall temperature		Skin friction coefficient	
	Nabwey et al. [45]	Current results	Nabwey et al. [45]	Current results
0.7	26.4590	26.459028	8.9609	8.960903
1	17.2876	17.287671	6.1409	6.140975
2	3.3635	3.363594	1.2489	1.248935

Fig. 2 Streamlines associated with the present fluid flow**Fig. 3** Influence of M on skin friction coefficient**Fig. 4** Influence of ϕ_1 on skin friction coefficient**Fig. 5** Influence of Da on skin friction coefficient**Fig. 6** Influence of E_{ck} on Nusselt number

interactions. As nanoparticle concentration grows, the fluid's effective viscosity increases, creating stronger resistance to flow near the boundary layer. Physically, this means higher nanoparticle loading makes the fluid “stickier” along surfaces, requiring more energy to maintain flow, thus elevating skin friction. This phenomenon is critical in applications like nanofluid-cooled systems, where optimizing particle concentration balances heat transfer enhancement against increased drag. It is noticed that the friction factor decreases by 2.5% (in the case of hybrid nanofluid) and 2.19% (in the case of mono nanofluid) when the value of M ranges from 0 to 3. It is evident from Figs. 6 and 7 that the rise in the Eckert number and Dufour number leads to the decrement in the Nusselt number. When Du rises, the

Fig. 7 Influence of Du on Nusselt number**Fig. 8** Influence of Sr on Sherwood number**Fig. 9** Influence of Kr on Sherwood number

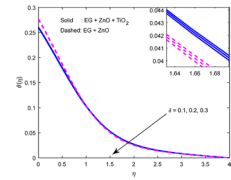
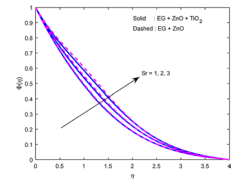
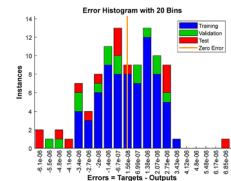
contribution of mass diffusion to heat transfer becomes more significant, creating a counter-acting thermal flux that opposes the primary temperature-driven heat transfer. This reduces the net temperature gradient near the boundary layer, weakening convective heat transfer efficiency. Consequently, the thermal boundary layer thickens, lowering the temperature gradient at the wall, which directly reduces the Nusselt number. The practical implication of the Dufour effect reducing the Nusselt number is significant in thermal management systems where simultaneous heat and mass transfer occur, such as chemical reactors, desalination plants, and electronics cooling with nanofluids. A weaker Nusselt number implies reduced convective heat transfer efficiency, meaning systems may require larger surface areas, enhanced cooling mechanisms, or adjusted operating conditions to compensate for the diminished thermal performance. Engineers must account for this coupling effect in designs—particularly in high-concentration-gradient environments—to avoid overheating or inefficiencies. It is noticed that the Nusselt number decreases by 39.7% (in case of hybrid nanofluid) and 38.5% (in case of mono nanofluid) when the value of Du ranges from 0 to 3. It is discovered that the rate of mass transfer reduced as Sr rises, whereas the same increases with the rise in Kr (see Figs. 8 and 9). When Sr increases, thermal diffusion dominates, causing solute particles to accumulate in colder zones, reducing concentration gradients near the surface. Since the Sherwood number represents the mass transfer rate relative to diffusion, a weaker concentration gradient leads to lower Sh values. Physically, this means that temperature-driven diffusion (Soret effect) competes with and partially suppresses concentration-driven mass transfer, diminishing the overall efficiency of species transport. This phenomenon is particularly relevant in non-isothermal flows, where thermal and solutal buoyancy forces interact. It is noticed that the Sherwood number decreases by 2.7% (in the case of hybrid nanofluid) and 2.97% (in the case of mono nanofluid) when the value of Sr ranges from 0 to 3.

Fig. 10 Impact of ϕ_1 on velocity profile**Fig. 11** Impact of Du on temperature profile

Regular Profiles Including Velocity

The rise in the volume fraction of nanoparticles reduces fluid velocity primarily due to increased viscosity and particle-fluid interactions. As nanoparticle concentration increases, the effective viscosity of the nanofluid rises, enhancing internal friction and resistance to flow. Additionally, Brownian motion and particle clustering disrupt smooth fluid layers, increasing drag forces. This results in a thicker momentum boundary layer and slower flow (see Fig. 10).

In thermal management systems, such as microchannel heat sinks or nuclear reactor coolants, excessive nanoparticle loading can impede flow rates, reducing heat transfer efficiency despite improved thermal conductivity—requiring a careful balance between particle concentration and pumping power. The rise in the Dufour number (Du) increases fluid temperature due to cross-diffusion effects, where concentration gradients induce additional heat flux in the system (see Fig. 11). As Du grows, mass diffusion (from concentration differences) generates extra thermal energy, effectively contributing to heat transfer alongside conduction and convection. This phenomenon enhances the thermal boundary layer, raising the fluid's temperature. In energy systems (e.g., solar collectors, nuclear reactors), a higher Du can improve heat transfer efficiency by leveraging concentration gradients. However, in microfluidic devices or chemical reactors, excessive Du may cause unintended overheating, requiring precise control of solute concentrations. The effect is also critical in geothermal and biomedical applications, where coupled heat-mass transport influences performance. The rise in the thermal relaxation parameter (associated with non-Fourier heat conduction models like Cattaneo-Christov) leads to a temperature decrement because it introduces a time lag between the applied heat flux and the temperature response (see Fig. 12). Essentially, the fluid requires more time to absorb and distribute thermal energy as the relaxation parameter increases, resulting in slower heat diffusion and lower overall temperature. This result is critical in high-speed thermal systems (e.g., hypersonic aircraft, laser processing, or electronics cooling), where traditional Fourier heat conduction models overpredict temperatures. By accounting for thermal relaxation, engineers can design more accurate cooling strategies for transient heat loads, preventing material degradation or failure. The increase in the Soret number (thermophoresis effect) enhances fluid concentration because it drives nanoparticles from hotter to cooler regions due to temperature-induced molecular motion

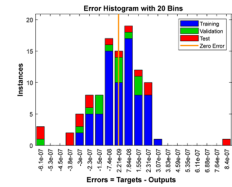
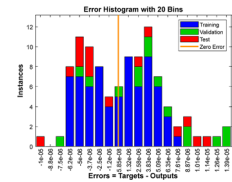
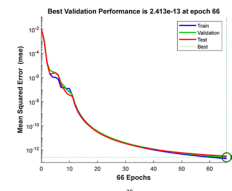
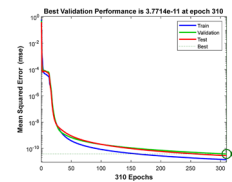
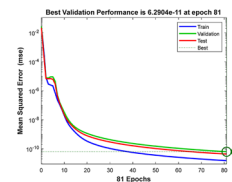
Fig. 12 Impact of δ on temperature profile**Fig. 13** Impact of Sr on concentration profile**Fig. 14** Error histogram for magnetic field parameter vs. skin friction coefficient

(see Fig. 13). As the Soret number rises, thermal diffusion becomes more dominant over molecular diffusion, causing particles to accumulate in colder zones.

Artificial Neural Network (ANN) Model

The use of Artificial Neural Networks (ANNs) to validate deterministic numerical simulations is justified here for several reasons. First, ANNs can serve as a surrogate model to rapidly predict results across a wide parameter space, reducing computational costs compared to repeated numerical simulations. Second, they provide an independent verification mechanism, helping detect potential biases or errors in the numerical scheme that might arise from discretization, convergence criteria, or implementation. Third, ANNs can interpolate and generalize from limited simulation data, identifying trends or anomalies that may not be immediately apparent from the numerical results alone. This capability is particularly useful in complex nonlinear systems where small perturbations could lead to significant deviations. Finally, integrating ANNs enhances the study's robustness by introducing a data-driven cross-validation approach, ensuring that the numerical solutions align not only with theoretical expectations but also with learned patterns from empirical or synthetic datasets. However, the limitations of ANNs—such as their reliance on training data quality and potential overfitting—must be acknowledged and mitigated through proper hyperparameter tuning and cross-validation.

The error histograms (Figs. 14 and 15, and 16) demonstrate the distribution of prediction errors, confirming whether residuals are centered near zero, indicating minimal bias. The performance plots (Figs. 17 and 18, and 19) track training progress, highlighting the best validation point, which ensures the model generalizes well without overfitting. The regression plots (Figs. 20 and 21, and 22) validate prediction accuracy by showing strong

Fig. 15 Error histogram for Dufour number vs. Nusselt number**Fig. 16** Error histogram for Soret number vs. Sherwood number**Fig. 17** Mean squared error (MSE) vs. training epoch for magnetic field parameter vs. skin friction coefficient**Fig. 18** Mean squared error (MSE) vs. training epoch for Dufour number vs. Nusselt number**Fig. 19** Mean squared error (MSE) vs. training epoch for Soret number vs. Sherwood number

correlations ($R \approx 1$) between targets and outputs. Finally, the function fit plots (Figs. 23 and 24, and 25) visually assess how well the model approximates the underlying data trends.

Conclusions

In the present study, we investigated the hybrid nanofluid flow over a permeable sphere in the presence of thermal radiation, non-Fourier heat flux, Soret, Dufour and chemical reaction parameters. The following are some of the study's most important findings:

Fig. 20 Regression chart for magnetic field parameter vs. skin friction coefficient

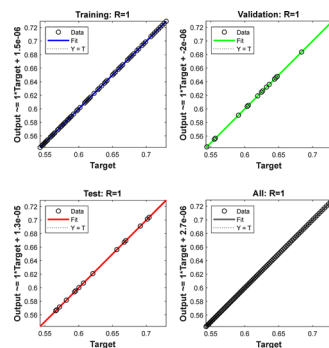


Fig. 21 Regression chart for Dufour number vs. Nusselt number

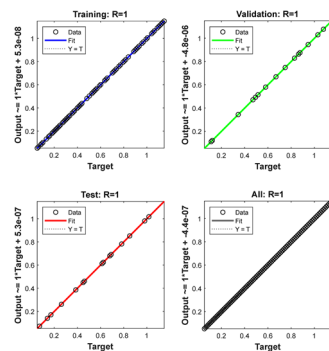


Fig. 22 Regression chart for Soret number vs. Sherwood number

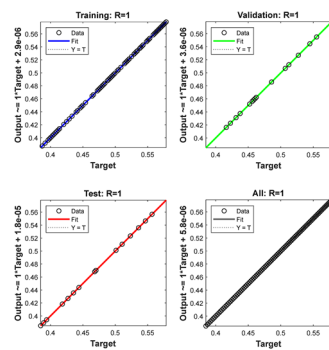


Fig. 23 Function fit plot for magnetic field parameter vs. skin friction coefficient

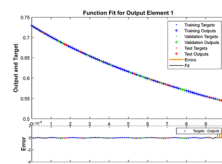
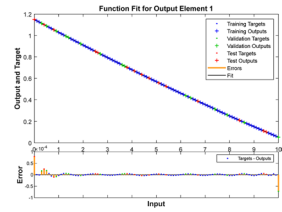
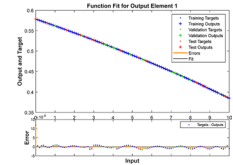


Fig. 24 Function fit plot for Dufour number vs. Nusselt number**Fig. 25** Function fit plot for Soret number vs. Sherwood number

- It is noticed that the friction factor decreases by 2.5% (in the case of hybrid nanofluid) and 2.19% (in the case of mono nanofluid) when the value of magnetic field parameter ranges from 0 to 3.
- It is observed that the Nusselt number decreases by 39.7% (in the case of hybrid nanofluid) and 38.5% (in the case of mono nanofluid) when the value of Dufour number ranges from 0 to 3.
- It is detected that the Sherwood number decreases by 2.7% (in the case of hybrid nanofluid) and 2.97% (in the case of mono nanofluid) when the value of Soret number ranges from 0 to 3.
- It is noticed that there is a decline in the temperature of the fluid with the rise in the thermal relaxation parameter.
- When the volume fraction of nanoparticles grown, the hike in fluid viscosity is noticed, obstructing the flow of fluid. As a result, the velocity drops.

Author Contributions Author-4 seen the methodology and numerical technique, results and graphs. Authors-1, 2, 3 seen the grammar, figures and tables. All Authors reviewed the manuscript.

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Declarations

Conflict of Interest The authors declare no competing interests.

References

1. Christov, C.I.: On frame indifferent formulation of the Maxwell–Cattaneo model of finite-speed heat conduction. *Mech. Res. Commun.* **36**(4), 481–486 (2009)
2. Straughan, B.: Thermal convection with the Cattaneo–Christov model. *Int. J. Heat Mass Transf.* **53**(1–3), 95–98 (2010)

3. Han, S., Zheng, L., Li, C., Zhang, X.: Coupled flow and heat transfer in viscoelastic fluid with Cattaneo–Christov heat flux model. *Appl. Math. Lett.* **38**, 87–93 (2014)
4. Li, J., Zheng, L., Liu, L.: MHD viscoelastic flow and heat transfer over a vertical stretching sheet with Cattaneo–Christov heat flux effects. *J. Mol. Liq.* **221**, 19–25 (2016)
5. Malik, M.Y., Khan, M., Salahuddin, T., Khan, I.: Variable viscosity and MHD flow in Casson fluid with Cattaneo–Christov heat flux model: Using Keller box method. *Eng. Sci. Technol. Int. J.* **19**(4), 1985–1992 (2016)
6. Dogonchi, A.S., Ganji, D.D.: Impact of Cattaneo–Christov heat flux on MHD nanofluid flow and heat transfer between parallel plates considering thermal radiation effect. *J. Taiwan Inst. Chem. Eng.* **80**, 52–63 (2017)
7. Dogonchi, A.S., Ganji, D.D.: Effect of Cattaneo–Christov heat flux on buoyancy MHD nanofluid flow and heat transfer over a stretching sheet in the presence of joule heating and thermal radiation impacts. *Indian J. Phys.* **92**(6), 757–766 (2018)
8. Ramandeivi, B., Reddy, J.R., Sugunamma, V., Sandeep, N.: Combined influence of viscous dissipation and non-uniform heat source/sink on MHD non-Newtonian fluid flow with Cattaneo–Christov heat flux. *Alexandria Eng. J.* **57**(2), 1009–1018 (2018)
9. Li, X., Khan, A.U., Khan, M.R., Nadeem, S., Khan, S.U.: Oblique stagnation point flow of nanofluids over stretching/shrinking sheet with Cattaneo–Christov heat flux model: Existence of dual solution. *Symmetry*. **11**(9), 1070 (2019)
10. Rana, S., Nawaz, M.: Investigation of enhancement of heat transfer in sutterby nanofluid using Koo–Kleinstreuer and Li (KKL) correlations and Cattaneo–Christov heat flux model. *Phys. Scr.* **94**(11), 115213 (2019)
11. Bhattacharyya, A., Seth, G.S., Kumar, R., Chamkha, A.J.: Simulation of Cattaneo–Christov heat flux on the flow of single and multi-walled carbon nanotubes between two stretchable coaxial rotating disks. *J. Therm. Anal. Calorim.* **139**(3), 1655–1670 (2020)
12. Khan, M.I., Alzahrani, F.: Transportation of heat through Cattaneo–Christov heat flux model in non-Newtonian fluid subject to internal resistance of particles. *Appl. Math. Mech.* **41**(8), 1157–1166 (2020)
13. Ahmad, S., Nadeem, S.: Flow analysis by Cattaneo–Christov heat flux in the presence of Thomson and troian slip condition. *Appl. Nanosci.* **10**(12), 4673–4687 (2020)
14. Reddy, M.G., Kumar, K.G.: Cattaneo–Christov heat flux feature on carbon nanotubes filled with micropolar liquid over A melting surface: A stream line study. *Int. Commun. Heat Mass Transfer*. **122**, 105142 (2021)
15. Sultana, U., Mushtaq, M., Muhammad, T., Albakri, A.: On Cattaneo–Christov heat flux in carbon-water nanofluid flow due to stretchable rotating disk through porous media. *Alexandria Eng. J.* **61**(5), 3463–3474 (2022)
16. Ali, B., Thumma, T., Habib, D., Riaz, S.: Finite element analysis on transient MHD 3D rotating flow of Maxwell and tangent hyperbolic nanofluid past a bidirectional stretching sheet with Cattaneo Christov heat flux model. *Therm. Sci. Eng. Progress*. **28**, 101089 (2022)
17. Moorthy, M.B.K., Senthilvadivu, K.: Soret and dufour effects on natural convection flow past a vertical surface in a porous medium with variable viscosity. *J. Appl. Math.* **2012**, 1–15 (2012)
18. Reddy, P.S., Chamkha, A.J.: Soret and dufour effects on MHD convective flow of Al₂O₃–water and TiO₂–water nanofluids past a stretching sheet in porous media with heat generation/absorption. *Adv. Powder Technol.* **27**(4), 1207–1218 (2016)
19. Mahdy, A., Ahmed, S.E.: Thermosolutal Marangoni boundary layer magnetohydrodynamic flow with the Soret and Dufour effects past a vertical flat plate. *Eng. Sci. Technol. Int. J.* **18**(1), 24–31 (2015)
20. Kefayati, G.R.: FDLBM simulation of entropy generation in double diffusive natural convection of power-law fluids in an enclosure with Soret and dufour effects. *Int. J. Heat Mass Transf.* **89**, 267–290 (2015)
21. Ramzan, M., Bilal, M., Chung, J.D.: Soret and Dufour effects on three dimensional upper-convected Maxwell fluid with chemical reaction and non-linear radiative heat flux. *Int. J. Chem. Reactor Eng.* **15**(3), (2017)
22. Farooq, A., Ali, R., Benim, A.C.: Soret and dufour effects on three dimensional Oldroyd-B fluid. *Phys. A: Stat. Mech. Its Appl.* **503**, 345–354 (2018)
23. Shojaei, A., Amiri, A.J., Ardahaie, S.S., Hosseinzadeh, K., Ganji, D.D.: Hydrothermal analysis of Non-Newtonian second grade fluid flow on radiative stretching cylinder with Soret and dufour effects. *Case Stud. Therm. Eng.* **13**, 100384 (2019)
24. Idowu, A.S., Falodun, B.O.: Soret–Dufour effects on MHD heat and mass transfer of walter’sb viscoelastic fluid over a semi-infinite vertical plate: Spectral relaxation analysis. *J. Taibah Univ. Sci.* **13**(1), 49–62 (2019)
25. Jha, B.K., Gambo, Y.Y.: Unsteady free convection And mass transfer flow past An impulsively started vertical plate with Soret And dufour effects: An Analytical approach. *SN Appl. Sci.* **1**(10), 1–11 (2019)

26. Rasool, G., Shafiq, A., Baleanu, D.: Consequences of Soret–Dufour effects, thermal radiation, and binary chemical reaction on Darcy Forchheimer flow of nanofluids. *Symmetry*. **12**(9), 1421 (2020)
27. Rebhi, R., Mamou, M., Hadidi, N.: Onset of linear and nonlinear thermosolutal convection with Soret and dufour effects in a porous collector under a uniform magnetic field. *Fluids*. **6**(7), 243 (2021)
28. Meng, Y., Li, B.: On viscoelastic fluid in a vertical porous media channel with Soret and dufour effects. *Appl. Math. Lett.* **124**, 107656 (2022)
29. Seid, E., Haile, E., Walelign, T.: Multiple slip, Soret and dufour effects in fluid flow near a vertical stretching sheet in the presence of magnetic nanoparticles. *Int. J. Thermofluids*, **100136**. (2022)
30. Arif, U., Nawaz, M., Selmi, A.L.: Numerical study of simultaneous transport of heat and mass transfer in Maxwell hybrid nanofluid in the presence of Soret and dufour effects. *Phys. Scr.* **97**, 025207 (2022)
31. Akbar, T., Ahmed, K., Muhammad, T., Munir, S.: Physical characteristics of dufour and Soret effects on MHD mixed convection flow of williamson fluid past a nonlinear stretching porous curved surface. *Waves Random Complex. Media*, 1–18. (2022)
32. Shilpa, B., Leela, V., Prasannakumara, B.C., Nagabhushana, P.: Soret and dufour effects on MHD double-diffusive mixed convective heat and mass transfer of couple stress fluid in a channel formed by electrically conducting and non-conducting walls. *Waves Random Complex. Media*, 1–22. (2022)
33. Li, S., Khan, M.I., Rafiq, M., Abdelmohsen, S.A., Abdullaev, S.S., Amjad, M.S.: Optimized framework for Darcy–Forchheimer flow with chemical reaction in the presence of Soret and dufour effects: A shooting technique. *Chem. Phys. Lett.* **825**, 140578 (2023)
34. Shilpa, B., Leela, V.: Galerkin finite element analysis of heat and mass transfer of jeffrey, Maxwell and Oldroyd-B nanofluids in a vertical annulus with an induced magnetic field and a non-uniform heat source/sink. *Int. J. Ambient Energy*. **44**(1), 1887–1903 (2023)
35. Shilpa, B., Leela, V.: An artificial intelligence model for heat and mass transfer in an inclined cylindrical annulus with heat generation/absorption and chemical reaction. *Int. Commun. Heat Mass Transfer*. **147**, 106956 (2023)
36. Shilpa, B., Badruddin, I.A., Reddy, R.G., Kamangar, S., Khan, A.A.: Exploration of linear and exponential heat source/sink with the significance of thermophoretic particle deposition on ZnO–SAE50 nano lubricant flow past a curved surface. *Case Stud. Therm. Eng.* **61**, 104883 (2024)
37. Rani, H.P., Shilpa, B., Leela, V., Reddy, R.G.: Numerical simulation and neural network model for hydromagnetic nanofluid convection in a porous wavy channel with thermal non-equilibrium model. *Phys. Scr.* **99**(11), 115219 (2024)
38. Shilpa, B., Leela, V., Badruddin, I.A., Kamangar, S., Bashir, M.N., Ali, M.M.: Integrated neural network based simulation of thermo solutal radiative double-diffusive convection of ternary hybrid nanofluid flow in an inclined annulus with thermophoretic particle deposition. *Case Stud. Therm. Eng.* **62**, 105158 (2024)
39. Shilpa, B., Leela, V., Badruddin, I.A., Kamangar, S., Ganesan, P., Khan, A.A.: Exploration of arrhenius activation energy and thermal radiation on MHD double-diffusive convection of ternary hybrid nanofluid flow over a vertical annulus with discrete heating. *Case Stud. Therm. Eng.* **65**, 105593 (2025)
40. Shilpa, B., Chohan, J.S., Beemkumar, N., Kulshreshtha, A., Ghodhban, R., Othman, N.A., Khan, M.I.: A novel machine learning approach for numerical simulation on the hybrid nanofluid flow past a converging/diverging channel: properties of tantalum and alumina nanoparticles. *Partial Diff. Equ. Appl. Math.* **13**, 101063 (2025).
41. Leela, V., Shilpa, B., Badruddin, I.A., Kamangar, S., Bashir, M.N., Ali, M.M.: Computational modelling and thermophysical characterization of Kelvin–Voigt fluid flow: A neural network approach. *ZAMM-Journal Appl. Math. Mechanics/Zeitschrift Für Angewandte Math. Und Mechanik*. **105**(5), e70073 (2025)
42. Alam, M.M., Alim, M.A., Chowdhury, M.M.: Viscous dissipation effects on MHD natural convection flow over a sphere in the presence of heat generation. *Nonlinear Analysis: Modelling Control*. **12**(4), 447–459 (2007)
43. Rao, D.P.C., Thiagarajan, S., Srinivas Kumar, V.: Significance of dufour and Soret effects on the non-Darcy flow of cross fluid by a Tilted plate with radiation and chemical reaction: A Cattaneo–Christov heat flux model. *Heat. Transf.* **51**(2), 1585–1600 (2022)
44. Jamshed, W., Aziz, A.: Cattaneo–Christov based study of Casson hybrid nanofluid flow over a stretching surface with entropy generation. *Appl. Nanosci.* **8**(4), 685–698 (2018)
45. Nabwey, H.A., Rashad, A.M., EL-Hakim, A., Alshber, S.I.: Effectiveness of newtonian heating on Magneto-Free convective flow of Polar nanofluid across a solid sphere. *Fractal Fract.* **6**(2), 57 (2022)
46. Molla, M.M., Hossain, M.A., Taher, M.A.: Magnetohydrodynamic natural convection flow on a sphere with uniform heat flux in presence of heat generation. *Acta Mech.* **186**(1), 75–86 (2006)

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